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# Smart Greenhouse Monitoring System: An ESP32-Based Automated Environmental Monitoring and Control System

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**Abstract**—The Smart Greenhouse Monitoring System is an ESP32-based automated prototype designed to monitor and regulate important environmental conditions required for healthy plant growth inside a greenhouse. The system combines a DHT22 temperature and humidity sensor, a soil-moisture sensor, an LDR module, and an MH-RD rain sensor with an ESP32 DevKit. Sensor readings are processed continuously and compared with predefined threshold values. According to the measured conditions, the controller operates a 12V DC water pump for irrigation, a DC fan for ventilation, a 12V light for additional illumination, and an SG90 servo motor for automatic roof movement. A 16×2 I2C LCD provides local system information. The prototype is intended to reduce continuous manual supervision, avoid unnecessary water and electricity use, and demonstrate how embedded systems, sensing, automation, and agricultural requirements can be integrated into one affordable platform. The design is suitable for laboratory demonstration and can be extended in the future with wireless dashboards, cloud storage, renewable power, and additional agricultural sensors.

**Index Terms**—ESP32, smart greenhouse, DHT22, soil moisture, rain detection, relay control, automation, smart agriculture.

## I. INTRODUCTION AND PROJECT OVERVIEW

Greenhouse cultivation provides a controlled environment in which plants can be protected from unfavorable outdoor conditions. However, successful greenhouse operation still requires regular observation of temperature, humidity, soil moisture, light availability, and weather conditions. In a manually managed greenhouse, delayed irrigation may cause plants to dry out, excessive temperature may create heat stress, insufficient light may reduce growth, and sudden rainfall may affect ventilation or damage sensitive plants.

The Smart Greenhouse Monitoring System is proposed as a compact and low-cost solution for automating these repetitive monitoring and control tasks. The ESP32 DevKit acts as the central controller of the prototype. It receives readings from the DHT22 sensor, soil-moisture sensor, LDR module, and rain sensor, then compares those readings with programmed threshold values. When the soil becomes dry, the system can start irrigation automatically. When temperature becomes too high, the fan can provide ventilation. When the available light becomes insufficient, the 12V light can be activated. Rain

detection is also used to control the greenhouse roof through an SG90 servo motor.

The project also includes a 16×2 I2C LCD for local display and three relay modules for switching the higher-voltage actuator loads. A 12V 5A adapter provides the main power supply, while the LM2596 buck converter regulates the voltage required by low-voltage electronics. The complete prototype therefore combines sensing, decision-making, electrical switching, mechanical movement, and plant-care functions within one system.

The main purpose of the prototype is educational as well as practical. It demonstrates how microprocessors and microcontrollers can be applied to a real agricultural problem using inexpensive and commonly available components. Although the current model is small, its modular structure allows future expansion into a larger IoT-enabled smart greenhouse.

## II. RELATED WORK

Previous greenhouse-monitoring research has commonly combined temperature, humidity, and soil-moisture sensing with automatic irrigation and ventilation. Such systems demonstrate that environmental data can be used to operate pumps and fans without continuous human intervention, helping to maintain more stable growing conditions and reduce unnecessary water use [1].

ESP32-based agricultural systems have also been used because the controller provides sufficient processing capability together with built-in wireless communication. These systems can collect environmental data and support real-time monitoring through local or cloud-based dashboards [2]. Other automated greenhouse designs use relay-controlled pumps, fans, and lighting systems to maintain suitable conditions using threshold-based control [3].

Precision-farming studies further show that sensor-based irrigation decisions can improve the efficient use of water and other resources [4]. The proposed Smart Greenhouse Monitoring System follows the same general direction but combines temperature and humidity monitoring, soil-moisture-

based irrigation, automatic ventilation, artificial lighting, rainfall detection, local LCD display, and servo-controlled roof movement in one laboratory prototype.

TABLE I. COMPARISON WITH SELECTED RELATED WORK

| Ref. | Main Work / Focus                                                                 | Relation to Proposed System                                                                                |
|------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| [1]  | IoT greenhouse: temperature, humidity, soil moisture; irrigation and ventilation. | Similar sensing and automatic plant-care control; proposed system also adds rain-based roof control.       |
| [2]  | ESP32-based agricultural monitoring with dashboard connectivity.                  | Supports ESP32 environmental monitoring; proposed prototype emphasizes local automatic actuation.          |
| [3]  | Automated greenhouse using relay-controlled environmental devices.                | Similar relay control; proposed system integrates LCD, rain sensor, and servo roof in one design.          |
| [4]  | Precision farming with sensor-based irrigation decisions.                         | Supports resource-efficient irrigation; proposed system combines irrigation with ventilation and lighting. |

### III. PROJECT OBJECTIVES

The project is designed to demonstrate an integrated greenhouse automation system rather than a single sensor-based function. The major objectives are listed below:

- Continuously measure greenhouse temperature and relative humidity using the DHT22 sensor.
- Monitor soil moisture so that irrigation decisions can be made from the actual soil condition instead of a fixed manual schedule.
- Automatically operate the 12V DC water pump when the soil becomes dry and stop irrigation after sufficient moisture is restored.
- Maintain a more suitable internal temperature by activating the DC fan when the measured temperature exceeds the selected threshold.
- Measure available natural light with the LDR module and provide additional lighting when the light level is insufficient.
- Detect rainfall using the MH-RD rain sensor and automatically control the greenhouse roof with the SG90 servo motor.
- Display system information and status through the 16×2 I2C LCD for easy local observation.
- Use relay modules to isolate and safely switch the pump, fan, and 12V lighting loads from the ESP32 control signals.
- Reduce unnecessary consumption of water and electrical energy by operating actuators only when their corresponding conditions require them.
- Develop an affordable, understandable, and expandable prototype suitable for microcontroller laboratory demonstration and future smart-agriculture development.

### IV. SYSTEM ARCHITECTURE AND COMPONENTS

The ESP32 DevKit is the main processing and control unit. The DHT22, soil-moisture sensor, LDR module, and rain sensor operate as input devices. Three relay modules and the SG90 servo motor operate as control outputs. The relay modules switch the 12V water pump, fan, and light, while the servo motor provides mechanical roof movement. The LCD

communicates with the ESP32 through the I2C interface.

The main supply is provided by a 12V 5A adapter. The LM2596 buck converter reduces and regulates the voltage for low-voltage components. Breadboards and jumper wires are used to assemble the prototype and provide temporary electrical connections during development and testing.

TABLE II. MAIN HARDWARE AND DEVELOPMENT TOOLS

| Component            | Purpose                                                            |
|----------------------|--------------------------------------------------------------------|
| ESP32 DevKit         | Reads sensor data, executes threshold logic, and controls outputs. |
| DHT22                | Measures temperature and relative humidity.                        |
| Soil Moisture Sensor | Detects the moisture condition of the soil.                        |
| Rain Sensor          | Detects rainfall for automatic roof control.                       |
| LDR Module           | Measures available natural light.                                  |
| Relay Modules        | Switch pump, fan, and 12V light loads.                             |
| Water Pump           | Provides automatic irrigation.                                     |
| DC Fan               | Provides ventilation and temperature control.                      |
| 12V Light            | Provides artificial lighting when needed.                          |
| SG90 Servo           | Moves the greenhouse roof.                                         |
| 16×2 I2C LCD         | Displays local system information.                                 |
| LM2596               | Regulates the supply voltage.                                      |
| Arduino IDE          | Programming and code-upload environment.                           |

### V. WORKING PROCEDURE

1. The 12V adapter is connected and the LM2596 provides the regulated voltage required by the control electronics.
2. The ESP32 initializes the DHT22, soil-moisture sensor, LDR module, rain sensor, relay outputs, LCD, and SG90 servo motor.
3. The controller continuously collects temperature, humidity, soil-moisture, light, and rainfall information.
4. Each reading is compared with its programmed threshold or control condition.
5. If the soil becomes dry, the water-pump relay is activated. The pump stops after sufficient moisture is restored.
6. If greenhouse temperature rises above the selected limit, the fan relay is activated to improve ventilation.
7. If the LDR indicates insufficient natural light, the 12V light is switched on; it turns off when enough light becomes available.
8. When rain is detected, the ESP32 commands the SG90 servo to close the roof. When rain is no longer detected, the roof can return to its normal position.
9. The LCD displays system information and the complete sensing-and-control cycle repeats continuously.

### VI. SYSTEM DESIGN / DIAGRAM

The implemented prototype integrates the physical greenhouse and the external controller circuit. Fig. 1 summarizes the connection between the sensors, ESP32, relays, actuators, display, and power circuit, while Fig. 2 shows the completed Smart Greenhouse Monitoring System prototype.

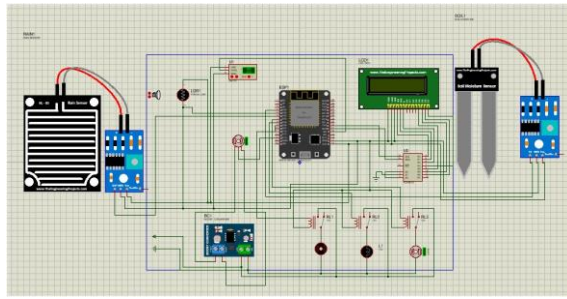


Fig. 1. Hardware connection and control-flow diagram.

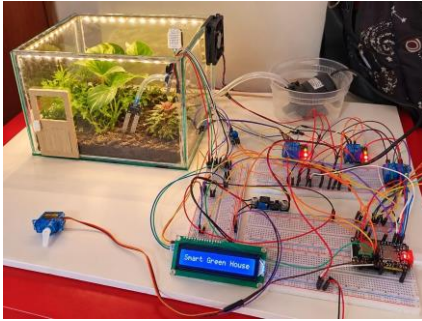


Fig. 2. Implemented Smart Greenhouse Monitoring System prototype.

## VII. EVALUATION PLAN

The prototype should be evaluated through repeated functional tests under different simulated greenhouse conditions. Each sensor should first be tested individually. The evaluation should confirm that the correct actuator responds when its corresponding environmental condition crosses the selected threshold.

For repeatability, each test should record the relevant condition, expected response, actual response, and whether the action occurred correctly. The response time of the water pump, fan, light, and servo motor can also be observed to determine whether the system reacts within an acceptable time. The DHT22 sensor can be tested under different temperature and humidity conditions, while the soil-moisture sensor can be tested with dry, moderately wet, and wet soil. Similarly, the LDR module can be tested under bright and low-light conditions, and the rain sensor can be tested using controlled water exposure.

The actuator tests should verify that the pump activates when soil moisture is low, the fan operates when temperature exceeds the selected threshold, the light turns on under insufficient illumination, and the servo closes the roof when rain is detected. The LCD should also be checked to confirm that system information is displayed correctly during operation.

To evaluate overall reliability, the complete system should be operated continuously for multiple test cycles while observing sensor readings, relay switching, servo movement, and power stability. Any false sensor reading, delayed response, unstable relay operation, or incorrect actuator action should be recorded and corrected through sensor calibration or threshold adjustment.

The final demonstration can report the results as successful

responses, failures, percentages, and average response times rather than relying on a single trial.

TABLE III. PROPOSED TEST METRICS

| Metric               | Measurement                                      |
|----------------------|--------------------------------------------------|
| Temperature response | Correct fan activations / high-temperature tests |
| Irrigation response  | Correct pump activations / dry-soil tests        |
| Lighting response    | Correct light activations / low-light tests      |
| Rain response        | Correct roof closures / rain-detection tests     |
| LCD response         | Successful status displays / trials              |
| Overall reliability  | Successful complete cycles / total runs          |

## VIII. BUDGET ANALYSIS

The initial proposal estimated the project cost at BDT 3,250. After selecting and purchasing the final components and greenhouse materials, the implemented prototype cost BDT 4,000. The final budget consists of BDT 2,830 for electronic components and BDT 1,170 for the greenhouse structure and supporting materials.

TABLE IV. ELECTRONIC COMPONENTS

| Electronic Component                | Cost (BDT) |
|-------------------------------------|------------|
| ESP32 DevKit                        | 450        |
| 12V 5A Adapter                      | 350        |
| LM2596 Buck Converter               | 85         |
| 1-Channel Relay Module (3 × 80)     | 240        |
| DHT22 Temperature & Humidity Sensor | 200        |
| Soil Moisture Sensor                | 50         |
| Rain Sensor (MH-RD)                 | 60         |
| LDR Module                          | 60         |
| SG90 Servo Motor                    | 140        |
| 12V DC Water Pump                   | 300        |
| 12V DC Fan                          | 100        |
| 12V Light                           | 100        |
| 16×2 I2C LCD                        | 255        |
| Jumper Wires                        | 100        |
| Breadboard (4 × 60)                 | 240        |
| USB Cable                           | 100        |
| Electronic Subtotal                 | 2,830      |

TABLE V. GREENHOUSE STRUCTURE / MATERIALS

| Material                       | Cost (BDT) |
|--------------------------------|------------|
| Glass                          | 850        |
| Transparent Acrylic Sheet      | 10         |
| PVC Board                      | 200        |
| Water Pipe - Pump              | 20         |
| Water Container                | 10         |
| Glue                           | 80         |
| Structure / Materials Subtotal | 1,170      |

**Final Project Cost = BDT 2,830 + BDT 1,170 = BDT 4,000.**

## IX. SUSTAINABILITY AND ENVIRONMENTAL IMPACT

The project combines computer science, electronics, agriculture, mechanical movement, and environmental management. Its modular construction allows damaged sensors

or control modules to be replaced individually instead of discarding the entire system. This improves technical and economic sustainability.

Automatic irrigation can reduce unnecessary water use because the pump operates only when the soil requires water. In the same way, the fan and light are operated according to environmental conditions instead of remaining on continuously. These features can reduce unnecessary consumption and support more efficient greenhouse management.

**Carbon Emission:** The system consumes electrical energy through the ESP32, sensors, pump, fan, and lighting. Therefore, indirect CO<sub>2</sub> emissions may occur depending on the source of electricity. Automated control can help reduce unnecessary energy consumption by operating devices only when required.

Possible negative impacts include electricity consumption and electronic waste after long-term use. Proper maintenance, reuse of working modules, responsible e-waste disposal, and future solar-power integration can reduce these impacts.

## X. SAFETY NORMS AND COMPLIANCE MEASURES

- Use insulated wires and secure all electrical connections.
- Keep the ESP32, relay modules, breadboards, and power connections away from water.
- Use the correctly rated 12V adapter and regulated voltage for low-voltage components.
- Keep the water container and irrigation pipe separated from exposed electronic circuits.
- Protect relay-controlled loads and avoid exposed conductors.
- Calibrate sensors before operation and inspect the wiring regularly.
- Ensure that the moving roof mechanism cannot injure users or damage plants.
- Disconnect the power supply before maintenance or circuit modification.

## XI. STAKEHOLDERS AND BENEFICIARIES

The possible beneficiaries include farmers and greenhouse owners, agricultural researchers, engineering students, universities and training institutions, smart-farming technology developers, government agricultural organizations, and environmental organizations. Students can use the prototype to learn microcontroller programming, environmental sensing, relay control, and automation, while larger versions of the same concept may support practical greenhouse operation.

## XII. SOCIETAL AND ECONOMIC ISSUES WITH MITIGATION STRATEGIES

The system may reduce routine manual work and unnecessary resource use, but practical implementation can also face cost, maintenance, power, and technical-knowledge

challenges. The principal issues and possible mitigation strategies are summarized below.

TABLE VI. PRINCIPAL ISSUES AND MITIGATIONS

| Issue                              | Mitigation                                                   |
|------------------------------------|--------------------------------------------------------------|
| Initial installation cost          | Use low-cost, locally available components.                  |
| Sensor failure / incorrect reading | Calibrate sensors and inspect them regularly.                |
| Power interruption                 | Add rechargeable battery or UPS backup in future.            |
| Water leakage                      | Physically separate water lines from electronics.            |
| Limited technical knowledge        | Provide simple instructions and user training.               |
| Electronic waste                   | Reuse working modules and dispose of damaged parts properly. |

## XIII. LIMITATIONS AND FUTURE WORK

The current system is a small laboratory prototype. Low-cost sensors may have limited accuracy, and the SG90 servo motor is suitable only for a lightweight roof. The prototype also depends on a stable power supply and does not yet store long-term environmental data.

- Add a Wi-Fi-enabled mobile or web dashboard and cloud data storage.
- Integrate solar power and rechargeable battery backup.
- Add pH, CO<sub>2</sub>, nutrient, and water-level sensors.
- Add notification alerts and graphical reports.
- Use a stronger motor mechanism for a larger greenhouse roof.
- Add manual override control and support multiple greenhouse zones.

## XIV. CONCLUSION

The Smart Greenhouse Monitoring System demonstrates how an ESP32 microcontroller can integrate multiple environmental sensors and actuators into one automated agricultural prototype. The system monitors temperature, humidity, soil moisture, light, and rainfall, then controls irrigation, ventilation, artificial lighting, and roof movement according to programmed conditions.

With a final implementation cost of approximately BDT 4,000, the project provides an affordable educational platform for studying microcontrollers, sensing, control logic, and smart-agriculture applications. Its modular design also provides a foundation for future development with wireless monitoring, additional sensors, renewable energy, and larger greenhouse structures.

## REFERENCES

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